223 results for “survey review wildfire ”forest fire” “machine learning” ”artificial intelligence” ”deep learning” techniques management”

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Predicting Vaping Cessation in Young Adults: A Machine Learning and Explainable Artificial Intelligence (XAI) Approach to Public Health Intervention.

D. M. M. M. Poolakkad S Satheeshkumar | BS² Ian Lango | +6 authors | M. P. Supriya D. Mahajan

Medicine, Computer Science · medRxiv · 18 September 2025

TLDR Key predictors of successful vaping cessation included age, environmental triggers, vaping frequency, sex, and long-term behavioral outlook, which underscore the utility of **machine learning** in uncovering nuanced behavioral and contextual determinants of addiction and cessation outcomes. Expand

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Vision-based fire management system using autonomous unmanned aerial vehicles: a comprehensive survey

Sufyan Danish | Md. Jalil Piran | +5 authors | Hyeonjoon Moon

Environmental Science, Engineering · Artificial Intelligence Review · 18 November 2025

TLDR This comprehensive **survey** critically examines recent advancements in vision-based **fire management** systems enabled by autonomous UAVs and explores how baseline **deep learning** models, including convolutional neural networks, attention mechanisms, YOLO variants, generative adversarial networks and transformers, enhance UAV capabilities for **fire**-related tasks. Expand

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A Synergistic Approach to Wildfire Prevention and Management Using AI, ML, and 5G Technology in the United States

S. Okoro | Alexander López | Austine Unuriode | Environmental Science, Computer Science · Social Science Research Network · 27 February 2024

TLDR The results showed that developing new technology presents an opportunity to detect and manage **wildfires** proactively, and AI-enabled remote sensing and 5G-based active monitoring can enhance proactive **wildfire** detection and **management**. Expand

7 | [PDF] | arXiv | Save | Cite

Proactive Wildfire Detection and Management using AI, ML, and 5G Technology in the United States

Okoro C. Stanley | Lopez Alexander | Unuriode O. Austine

Environmental Science, Computer Science · International Journal of Data Mining &amp… · 29 March 2024

TLDR The results showed that developing new technology presents an opportunity to detect and manage **wildfires** proactively, and AI-enabled remote sensing and 5G-based active monitoring can enhance proactive **wildfire** detection and **management**. Expand

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Machine learning in infection management using routine electronic health records: tools, techniques, and reporting of future technologies.

C. Luz | M. Vollmer | J. Decruyenaere | M. Nijsten | C. Glasner | B. Sinha

Medicine, Computer Science · Clinical Microbiology and Infection · 12 February 2020

TLDR Clinicians and researchers are supported to navigate through the methodological aspects of ML approaches in the field of infection **management** to support building trust in these new technologies will require improved reporting. Expand

84 PDF PubMed Save Cite

Enterprise generative artificial intelligence technologies, Internet of Things and blockchain-based fintech management, and digital twin industrial metaverse in the cognitive algorithmic economy

Tomas Kliestik Robert Dragomir +14 authors Dominika Gajdosikova

Computer Science, Business · Oeconomia Copernicana · 30 December 2024

TLDR It is shown that enterprise generative AI-driven schedule prediction tools, job search and algorithmic hiring systems, and synthetic training data can improve team selection, job performance and firing decisions, hiring decision processes, and workforce productivity in terms of prediction and decision-making by use of algorithmic **management**, system performance, and production process tracking tools. Expand

33 PDF PDF Save Cite

Privacy-Preserving Data-Driven Learning Models for Emerging Communication Networks: A Comprehensive Survey

Mostafa M. Fouda Z. Fadlullah Mohamed I. Ibrahem Nei Kato

Computer Science, Engineering · IEEE Communications Surveys and Tutorials · 1 August 2025

TLDR A tutorial-oriented **review** of data-driven models, which can be potentially mapped to privacy-preserving **techniques**, and preliminaries of a number of privacy-preserving **techniques** that can be potentially adopted for emerging communication networks. Expand

61 IEEE Save Cite

Smart Water Management: Role of IoT, AI, and Machine Learning in Water Quality Monitoring

Papia Dutta S. Sarma Environmental Science, Computer Science · Water environment research · 1 February 2026

TLDR This **review** synthesizes the latest progress in IoT-, AI-, and ML-driven water quality monitoring, outlines quantitative improvements reported in recent literature, and highlights remaining technical, economic, and governance challenges influencing large-scale deployment. Expand

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An understanding of machine learning techniques in big data analytics: a survey

S. Isabella S. Srinivasan Computer Science · International journal of engineering and… · 8 June 2018

TLDR This study **reviews** the various challenges and innovative ideas for big data analytics with **machine learning** in different fields over the past ten years and identifies the research projects based on the discussions over **machine learning techniques** forbig data analytics and provides suggestions to develop the new projects. Expand

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Forest Fire Detection, Classification, and Segmentation Using Satellite Imagery: A Case Study in the Amazon Region

A. Mahmoud Environmental Science, Computer Science · Stardom Scientific Journals of Natural and… · 30 July 2025

TLDR The results showed that the integration of optical and thermal data significantly improved early detection efficiency, and the use of **machine learning** algorithms increased classification accuracy to over 85%. Expand

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